



IngeniumAI

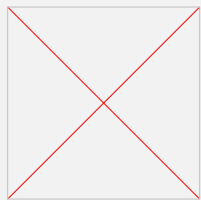
Early disease detection
with no extra scans

Jeff Clark

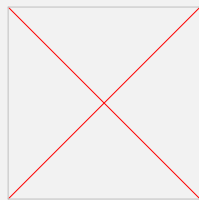
Co-founder & CEO (+ Bristol AI Research Fellow)

Mission statement

We aim to **reduce time-to-diagnosis for uncommon diseases by at least 80%** by analysing routinely collected medical imaging data.



Over 75%
of all patient care
involves radiology

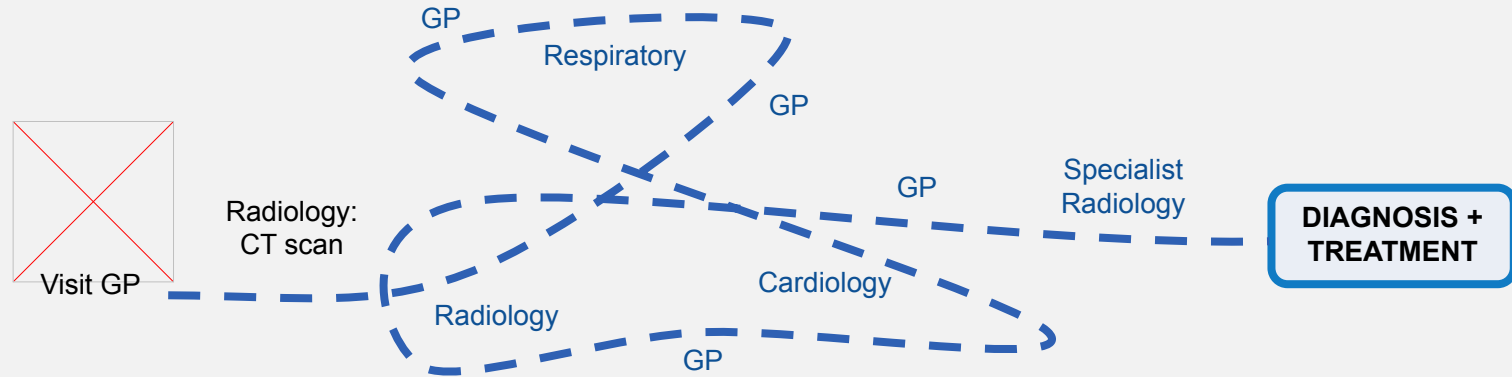


40% shortage
in NHS radiologists
by 2028



Most radiologists are not
equipped to spot many
deadly conditions

The problem



Application 1*: Pulmonary Hypertension (PH)

Deadly disease, treatable if caught early

- Vague symptoms, low prevalence, lack of specialists
- Estimated 3000 unnecessary deaths in UK and £50M annual cost to NHS
- Visible signs of disease in routine medical imaging

Average time to diagnosis: **2 years, 2 months**
Average time to mortality: **2 years, 6 months**

*Other applications under consideration

European Heart Journal

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Volume 43, Issue 38

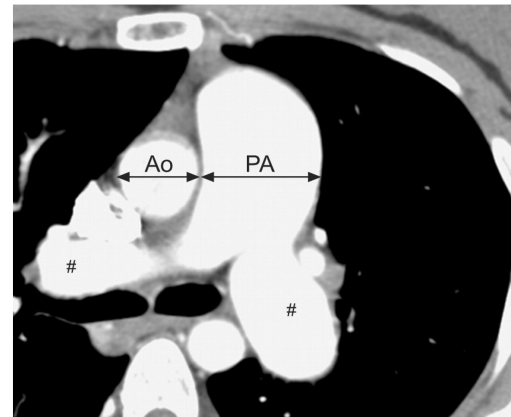
7 October 2022

Article Contents

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JOURNAL ARTICLE GUIDELINES

2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension: Developed by the task force for the diagnosis and treatment of pulmonary hypertension of the European Society of Cardiology (ESC) and the European Respiratory Society (ERS). Endorsed by the International Society for Heart and Lung Transplantation (ISHLT) and the European Reference Network on rare respiratory diseases (ERN-LUNG). 



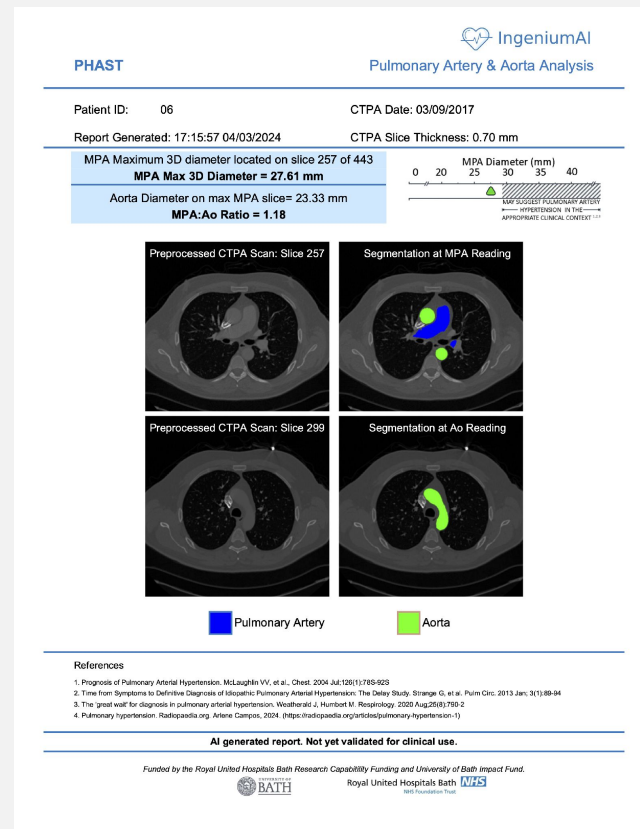
Our solution PHAST[®]

Pulmonary Hypertension Automated Screening Technology
SaMD, working towards class IIa certification



- ✓ Follows existing clinical guidelines
- ✓ Bring specialist radiology expertise to every clinic
- ✓ Streamlined referrals
 - ✓ Reduce healthcare system burden
 - ✓ Faster and more effective patient treatment

Diagnosis made in weeks not years



Example patient report

AI Hackathon Problem Statement

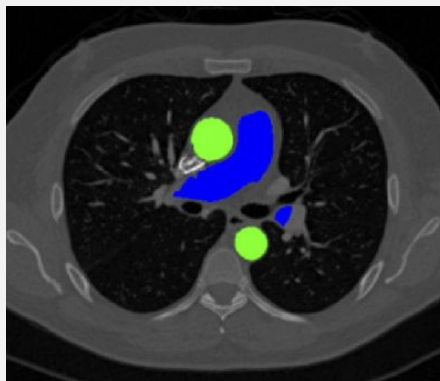
Anatomical segmentation from **non-contrast** chest CT scans

- Currently, PHAST is trained to work on contrast CT (CTPA) scans
- Non-contrast CT capability would allow detection earlier with incidental findings, in those who are allergic to contrast, and in emergency settings

Contrast CT



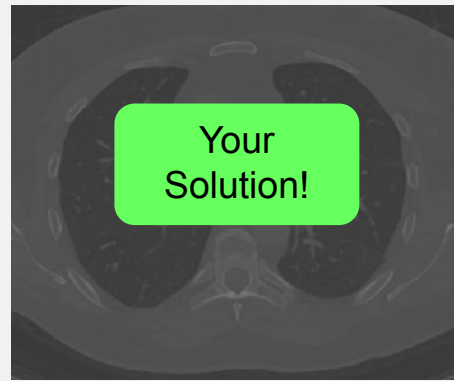
PHAST Segmentation



Non-contrast CT



Segmentation



What we're looking for

- Proof-of-concept demonstrating the feasibility of anatomical segmentation on non-contrast CT scans
 - Qualitatively
 - Quantitatively
- Presentation which clearly explains design choices, assumptions, limitations, and outline how the approach could be improved or scaled with further development

What we're providing

- Contrast and non-contrast CT scans
- Anatomical labels for the aorta and pulmonary artery
- Trained segmentation model (using contrast scans)
- Python notebook to get you started
- Test set, to be released 11am on Friday

Prizes for the winning team

- Gift vouchers and certificates
- Opportunity to present to our board of directors and discuss next steps



IngeniumAI

Any questions?



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